STAT5080 – HW 6 Solutions

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Date: 3/12/26

Problem 1

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## trt          4  561.9  140.47   9.931 5.94e-05 ***
## Residuals   25  353.6   14.14
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Treatment means: 13.8 14 8 17.2 22.4
## Grand mean:      14.8667
## SSE = 353.6
## MSE = 14.144
## dfe = 25
```

Part (a): 95% Upper Confidence Limit for σ^2

A 100(1-α)% upper confidence limit for σ^2 is:

$$\left(0,\ \frac{SSE}{\chi^2_{N-a,\,\alpha}}\right)$$

```
## SSE = 353.6
## dfe = 25
## chi-sq(alpha=0.05, df= 25 ) = 14.6114
## 95% Upper Confidence Limit for sigma^2: 24.2003
```

Conclusion: The 95% upper confidence limit for σ^2 is 24.2003

Part (b): Sample Size to Detect Maximum Difference of 5

We wish to detect $\Delta \geq 5$ with power ≥ 0.95 at $\alpha = 0.05$.

Using $\hat{\sigma}^2 =$ UCL from part (a), and the non-centrality parameter:

$$\lambda = \frac{n\Delta^2}{2\sigma^2}$$

```
##     n  Fcrit  lambda   beta  power
## 1   2 5.1922  1.0330 0.9244 0.0756
## 2   3 3.4780  1.5496 0.8934 0.1066
## 3   4 3.0556  2.0661 0.8598 0.1402
## 4   5 2.8661  2.5826 0.8241 0.1759
## 5   6 2.7587  3.0991 0.7867 0.2133
## 6   7 2.6896  3.6157 0.7481 0.2519
## 7   8 2.6415  4.1322 0.7088 0.2912
## 8   9 2.6060  4.6487 0.6692 0.3308
```

```
## 9  10 2.5787  5.1652 0.6297 0.3703
## 10 11 2.5572  5.6818 0.5906 0.4094
## 11 12 2.5397  6.1983 0.5523 0.4477
## 12 13 2.5252  6.7148 0.5148 0.4852
## 13 14 2.5130  7.2313 0.4786 0.5214
## 14 15 2.5027  7.7478 0.4437 0.5563
## 15 16 2.4937  8.2644 0.4102 0.5898
## 16 17 2.4859  8.7809 0.3783 0.6217
## 17 18 2.4790  9.2974 0.3480 0.6520
## 18 19 2.4729  9.8139 0.3194 0.6806
## 19 20 2.4675 10.3305 0.2925 0.7075
## 20 21 2.4626 10.8470 0.2673 0.7327
## 21 22 2.4582 11.3635 0.2437 0.7563
## 22 23 2.4542 11.8800 0.2218 0.7782
## 23 24 2.4506 12.3966 0.2015 0.7985
## 24 25 2.4472 12.9131 0.1826 0.8174
## 25 26 2.4442 13.4296 0.1653 0.8347
## 26 27 2.4414 13.9461 0.1493 0.8507
## 27 28 2.4387 14.4626 0.1346 0.8654
## 28 29 2.4363 14.9792 0.1212 0.8788
## 29 30 2.4341 15.4957 0.1089 0.8911
##
## Minimum sample size per treatment: 37
## Achieved power: 0.9505
```

Conclusion: We need $n = 37$ observations per treatment to detect a maximum difference of 5 with probability at least 0.95.

Part (c): Sample Size via Bonferroni CI

We want ± 5 accuracy for a 95% CI on $\mu_i - \mu_j$ using Bonferroni's method.
Number of pairwise comparisons: $m = \binom{5}{2} = 10$.

Bonferroni half-width: $t_{N-a,\,1-\alpha/(2m)}\sqrt{\frac{2 \cdot MSE}{n}}$

We search for the smallest n such that this half-width ≤ 5.

```
##     n dfe  t_bon half_width
## 7   8  35 2.9960     7.3693
## 8   9  40 2.9712     6.8902
## 9  10  45 2.9521     6.4946
## 10 11  50 2.9370     6.1607
## 11 12  55 2.9247     5.8738
## 12 13  60 2.9146     5.6237
## 13 14  65 2.9060     5.4033
## 14 15  70 2.8987     5.2070
## 15 16  75 2.8924     5.0307
## 16 17  80 2.8870     4.8713
##
## Minimum n per treatment (Bonferroni): 17
## Half-width achieved: 4.8713
```

Conclusion (Bonferroni): We require a sample size of $n = 17$ rats per treatment to ensure that the 95% confidence intervals for all pairwise differences in mean activity scores have an accuracy of at least ± 5.

Part (d): Sample Size via Tukey's CI

The half-width of the Tukey interval is given by:

$$E = \frac{q_\alpha(a, f)}{\sqrt{2}} \sqrt{\frac{2 \cdot \sigma^2}{n}}$$

Where:

- $q_\alpha(a, f)$ is the upper α percentage point of the studentized range distribution.
- $f = a(n-1)$ is the degrees of freedom for error.

```
##     n dfe  q_tuk half_width
## 6   7  30 2.9006     7.6272
## 7   8  35 2.8751     7.0718
## 8   9  40 2.8561     6.6233
## 9  10  45 2.8414     6.2512
## 10 11  50 2.8298     5.9359
## 11 12  55 2.8203     5.6641
## 12 13  60 2.8125     5.4267
## 13 14  65 2.8058     5.2170
## 14 15  70 2.8002     5.0299
## 15 16  75 2.7953     4.8617
##
## Minimum n per treatment (Tukey): 16
## Half-width achieved: 4.8617
```

Conclusion (Tukey): To achieve an accuracy of ± 5 for all pairwise mean activity score comparisons, we require a sample size of $n = 16$ rats per treatment. This is slightly more efficient than the Bonferroni method, as Tukey's procedure is specifically optimized for all-possible pairwise comparisons.

Part (e): Compare Bonferroni vs. Tukey Sample Sizes

```
## Bonferroni sample size: 17
## Tukey sample size:      16
## Difference:             1
```

Bonferroni requires n=17; Tukey's requires n=16

Tukey's gives a slightly smaller sample size, which makes sense: Tukey's method is designed specifically and optimally for all pairwise comparisons, guaranteeing exactly 100(1-α)% confidence. Bonferroni, by contrast, only guarantees the confidence level is *at least* 100(1-α))%, making it more conservative (wider CIs), and thus requiring more observations to achieve the same accuracy. For pairwise comparisons specifically, Tukey's is the sharper method, so it will always require equal or fewer observations than Bonferroni for the same accuracy target.

Problem 2

Five looms are chosen **at random** from a large population of looms.

```
## Loom means:
##     1     2     3     4     5
## 14.30 13.72 14.06 13.80 13.82
## Grand mean: 13.94
##
## Analysis of Variance Table
##
## Response: output
##           Df Sum Sq Mean Sq F value  Pr(>F)
## loom       4  1.132  0.2830   3.238 0.03345 *
## Residuals 20  1.748  0.0874
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## MST = 0.283
## MSE = 0.0874
## F0  = 3.238
## p-value = 0.0334
```

Part (a): Why Is This a Random Effects Experiment?

Answer: This is a random effects experiment because the five looms are chosen at random from a large population of looms in the textile mill. The goal is not to make inferences about these specific five looms, but rather to draw conclusions about the entire population of looms. The loom effects τ_i are treated as random variables drawn from a distribution $\tau_i \sim N(0, \sigma_\tau^2)$, rather than as fixed unknown constants.

Part (b): Are the Looms Equal in Output?

At the $\alpha = 0.05$ significance level, we tested the null hypothesis $H_0 : \sigma_\tau^2 = 0$ (all looms have equal mean output) against the alternative $H_1 : \sigma_\tau^2 > 0$ (at least one loom differs).

```
## F0 = 3.238
## F critical value F(0.95, 4, 20) = 2.8661
## p-value = 0.033449
## Conclusion: REJECT H0. There is significant evidence that looms differ in output.
```

Since our observed test statistic $F_0 = 3.238$ exceeds the critical value $F_{0.05,4,20} = 2.8661$, and our p-value $= 0.0334$ is less than 0.05, we have sufficient evidence to reject our H_0 to conclude that the variability between looms is significantly greater than the internal experimental error. In practical terms, the specific loom used has a statistically significant impact on the total output.

Part (c): Estimate the Variability Between Looms ($\hat{\sigma}_\tau^2$)

In the random effects model, $E(MST) = \sigma^2 + n\sigma_\tau^2$ and $E(MSE) = \sigma^2$.

Therefore: $\hat{\sigma}_\tau^2 = \frac{MST - MSE}{n}$

```
## Estimate of sigma^2 (error variance):       0.0874
## Estimate of sigma_tau^2 (loom variance):    0.03912
```

- **Error Variance (Within-group):**

$$\hat{\sigma}^2 = MSE = 0.0874$$

- **Loom Variance (Between-group):**

$$\hat{\sigma}_\tau^2 = \frac{MST - MSE}{n} = \frac{0.283 - 0.0874}{5} = 0.03912$$

Part (d): Estimate the Experimental Error Variance ($\hat{\sigma}^2$)

```
## Estimate of sigma^2 = MSE = 0.0874
```

Estimate: $\hat{\sigma}^2 = MSE = 0.0874$

Part (e): 95% CI for σ_τ^2 (Variability Between Looms)

Using the Satterthwaite approximation, the 95% confidence interval for the loom variance component σ_τ^2 is calculated as:

$$CI = \left(\frac{x \cdot \hat{\sigma}_\tau^2}{\chi^2_{\alpha/2,x}}, \frac{x \cdot \hat{\sigma}_\tau^2}{\chi^2_{1-\alpha/2,x}} \right)$$

```
## Satterthwaite df (x) = 1.8751
## chi2_{0.975, x}      = 7.1122
## chi2_{0.025, x}      = 0.0384
##
## 95% CI for sigma_tau^2: ( 0.010314 , 1.907764 )
```

The 95% confidence interval for σ_τ^2 is $[0.010314, 1.907764]$.

Since the 95% confidence interval for σ_τ^2 does not contain zero, we are 95% confident that there is a non-zero variance component attributable to the looms. This confirms that the differences in output between the individual looms are statistically significant and not merely due to random experimental error.

Part (f): 95% CI for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$

The Intraclass Correlation Coefficient (ICC) represents the proportion of total variance attributable to the differences between looms:

$$\rho = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2}$$

Using the F-distribution method with $df_1 = 4$ and $df_2 = 20$, the 95% confidence interval for the ICC is calculated as:

```
## F_{0.975, 4, 20} = 3.5147
## F_{0.025, 4, 20} = 0.1168
## F0 = MST/MSE     = 3.238
## L (sigma_tau2/sigma2) = -0.0157
## U (sigma_tau2/sigma2) = 5.3434
##
## 95% CI for sigma_tau^2 / (sigma_tau^2 + sigma^2):
## ( -0.016 , 0.8424 )
```

The 95% confidence interval for the ICC is $[-0.016, 0.8424]$.

While our previous point estimate showed a notable contribution from the looms, the wide interval, extending from near zero to over 80%, reflects a high degree of uncertainty in the exact proportion of variance, likely due to the small sample size or degrees of freedom.

Part (g): Is There Significant Variation from Loom to Loom?

From part (b), the ANOVA F-test evaluates $H_0 : \sigma_\tau^2 = 0$.

```
## F0 = 3.238 , p-value = 0.0334
## Alpha = 0.05
## Decision: REJECT H0
```

Since the p-value = 0.0334, we reject H_0. There is significant evidence that looms differ in output.

Part (h): Test $H_0 : \sigma_\tau^2/\sigma^2 \leq 1$ vs. $H_a : \sigma_\tau^2/\sigma^2 > 1$

This tests whether the between-loom variance is no larger than the within-loom (error) variance, i.e., $\gamma_0 = 1$.

Under $H_0 : \sigma_\tau^2/\sigma^2 = \gamma_0$, the test statistic is:

$$F^* = \frac{MST}{MSE(n\gamma_0 + 1)} \sim F_{a-1,N-a}$$

```
## gamma_0 = 1
## F* = MST / [MSE*(n*gamma_0 + 1)] = 0.5397
## F critical value F(0.95, 4, 20) = 2.8661
## p-value = 0.7084
## Decision: FAIL TO REJECT H0
```

At the $\alpha = 0.05$ significance level, we fail to reject the null hypothesis. The observed test statistic $F^* = 0.5397$ is well below the critical threshold of 2.8661, and the large p-value (0.7084) indicates that our data is highly consistent with the assumption that the loom variance component is approximately equal to the error variance. Practically, this suggests that the magnitude of the variability between different looms is not significantly different from the magnitude of the random noise within the production process.

Part (i): Sample Size Determination for Repeated Experiment

Parameters: $\gamma = 1$, $\alpha = 0.01$, desired power = 0.90, $\Delta = 3$ (i.e., $\sigma_\tau^2/\sigma^2 = 3$).

```
## === Finding n given a fixed number of looms ===
##
## --- v = 47 looms ---
##   Sample_Size  lambdas    Powers
## 1           5 4.000000 1.0000000
## 2           2 2.645751 0.9999808
## 3           3 3.162278 1.0000000
## 4           4 3.605551 1.0000000
##
## --- v = 5 looms (original experiment size) ---
##   Sample_Size  lambdas    Powers
## 1           5 4.000000 0.8894090
## 2           5 4.000000 0.8894090
## 3           6 4.358899 0.9248260
## 4           7 4.690416 0.9456153
##
## --- v = 10 looms ---
##   Sample_Size  lambdas    Powers
## 1           5 4.000000 0.9950752
## 2           2 2.645751 0.6940509
## 3           3 3.162278 0.9477264
## 4           4 3.605551 0.9861705
```

Case 1: Power Analysis for $H_0 : \sigma_\tau^2/\sigma^2 \leq 1$

For the specific requirement of $\gamma = 1$, the analysis confirms that $a = 47$ looms with $n = 3$ observations each (Total $N = 141$) is the correct requirement. At $n = 3$, the power is 1.000, which safely exceeds the 0.90 threshold.

Case 2: Power Analysis for $H_0 : \sigma_\tau^2 = 0$

The `sample_size_sigtau_2` function tests the simpler hypothesis that there is no loom variance at all. This results in smaller required sample sizes that are not directly applicable to the $\gamma = 1$ constraint:

- With $v = 5$ looms: Requires $n = 6$ observations per loom ($N = 30$) to reach a power of 0.925.
- With $v = 10$ looms: Requires $n = 3$ observations per loom ($N = 30$) to reach a power of 0.948.

Conclusion

The appropriate design for Problem 2(i) is $a = 47$ looms and $n = 3$ observations, totaling 141 observations.

While using 5 or 10 looms requires fewer total observations ($N = 30$) to detect the mere *presence* of loom variance ($H_0 : \sigma_\tau^2 = 0$), those designs lack the statistical power required to test the more stringent hypothesis that the variance ratio γ is less than or equal to 1. Therefore, Method 1 remains the valid approach for this specific problem.

Problem 3

3A. Verify the value of C

```
## [1] 3.264706
```

The constant C is used in the estimation of the variance component for the random effects model and is defined as

$$C = \frac{N^2 - \sum_{i=1}^{a} n_i^2}{N(a-1)},$$

where N is the total number of observations, a is the number of groups (heads), and n_i is the number of observations in the ith group. Using the observed sample sizes from the five heads, the computed value from the R output is $C = 3.264706$. This value verifies the correction factor required when estimating the head-to-head variance component under unequal sample sizes. The constant C essentially adjusts for the imbalance in the number of observations taken from each head and ensures that the variance component estimator for the head-to-head variability remains unbiased. Therefore, the R output confirms that the calculated value of C is approximately 3.2647, which will be used in estimating the variance component associated with differences among the machine heads.

3B. Is there evidence for head-to-head variability?

```
##             Df  Sum Sq  Mean Sq F value  Pr(>F)
## group        4 0.05766 0.014415   7.294 0.00322 **
## Residuals   12 0.02372 0.001976
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To determine whether there is evidence of variability among the machine heads, we perform a one-way ANOVA test under the random effects framework. The hypotheses are

$$H_0 : \sigma_\alpha^2 = 0$$

which states that there is no head-to-head variability (all heads fill bottles with the same mean weight), versus

$$H_1 : \sigma_\alpha^2 > 0,$$

which indicates that variability exists among the heads. From the R output, the ANOVA table gives an F statistic of $F = 7.294$ with numerator degrees of freedom 4 and denominator degrees of freedom 12, and an associated p-value of 0.00322. Since this p-value is much smaller than the common significance level of $\alpha = 0.05$, we reject the null hypothesis. This provides strong statistical evidence that the mean filling weights differ among the machine heads. In practical terms, this suggests that the heads on the filling machine are not operating identically, and some heads may systematically dispense slightly different amounts of oil compared to others.

3C. Estimate head-to-head and error variabilities.

```
## [1] 0.00197639
## [1] 0.003810016
```

The variance components for the random effects model can be estimated using the mean squares from the ANOVA table. The error variance is estimated directly by the residual mean square, while the head-to-head variance is estimated using the difference between the treatment mean square and the error mean square adjusted by the constant C. From the R output, the residual mean square (MSE) is 0.00197639, which provides the estimate of the error variance $\hat{\sigma}^2 = 0.00197639$. This represents the variability in bottle weights due to random measurement error or other within-head fluctuations. The estimated head-to-head variance component is $\hat{\sigma}_\alpha^2 = 0.003810016$, which reflects the variability attributable to differences among the filling heads. Since the estimated head-to-head variance is larger than the error variance, this indicates that a meaningful portion of the total variability in bottle weights arises from systematic differences between the machine heads rather than random measurement variation alone.

Problem 4

In the one-way random effects model:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, \ldots, a, \quad j = 1, 2, \ldots, n_i$$

where:

- $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$
- $\tau_i \overset{iid}{\sim} N(0, \sigma_\tau^2)$
- $\tau_i \perp \varepsilon_{ij}$ for all i, j
- $N = \sum_{i=1}^{a} n_i$ is the total number of observations

Part (a): Show that $E(\mathbf{MST}) = \sigma^2 + C\sigma_\tau^2$, where $C = \dfrac{N^2 - \sum_{i=1}^{a} n_i^2}{N(a-1)}$

The treatment sum of squares,

$$\text{SST} = \sum_{i=1}^{a} n_i \left(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}\right)^2$$

where the treatment mean and grand mean are:

$$\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}, \qquad \bar{y}_{\cdot\cdot} = \mu + \bar{\tau}_{\cdot\cdot} + \bar{\varepsilon}_{\cdot\cdot}$$

with $\bar{\tau}_{\cdot\cdot} = \dfrac{\sum_{i=1}^{a} n_i \tau_i}{N}$ and $\bar{\varepsilon}_{\cdot\cdot} = \dfrac{\sum_{i=1}^{a} \sum_{j=1}^{n_i} \varepsilon_{ij}}{N}$.

Therefore:

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = (\tau_i - \bar{\tau}_{\cdot\cdot}) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$$

Now,

$$E(\text{SST}) = \sum_{i=1}^{a} n_i \, E[(\tau_i - \bar{\tau}_{\cdot\cdot}) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})]^2$$

Since $\tau_i \perp \varepsilon_{ij}$, the cross-product terms vanish in expectation:

$$E(\text{SST}) = \underbrace{\sum_{i=1}^{a} n_i \, E(\tau_i - \bar{\tau}_{\cdot\cdot})^2}_{A} + \underbrace{\sum_{i=1}^{a} n_i \, E(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}_{B}$$

Term B,

Since $\bar{\varepsilon}_{i\cdot} \sim N(0,\, \sigma^2/n_i)$ and $\bar{\varepsilon}_{\cdot\cdot} \sim N(0,\, \sigma^2/N)$, this is the standard ANOVA result for fixed effects:

$$B = (a-1)\sigma^2$$

Term A,

$$A = \sum_{i=1}^{a} n_i \left[E(\tau_i^2) - 2E(\tau_i \bar{\tau}_{\cdot\cdot}) + E(\bar{\tau}_{\cdot\cdot}^2)\right]$$

Since τ_i are i.i.d. with $E(\tau_i) = 0$, $\text{Var}(\tau_i) = \sigma_\tau^2$, and $E(\tau_i \tau_k) = 0$ for $i \neq k$:

$$E(\tau_i^2) = \sigma_\tau^2$$

$$E(\tau_i \bar{\tau}_{\cdot\cdot}) = E\left(\tau_i \cdot \frac{\sum_{k=1}^{a} n_k \tau_k}{N}\right) = \frac{n_i}{N}\,\sigma_\tau^2$$

$$E(\bar{\tau}_{\cdot\cdot}^2) = E\left(\frac{\sum_{i=1}^{a} n_i \tau_i}{N}\right)^2 = \frac{\sum_{i=1}^{a} n_i^2}{N^2}\,\sigma_\tau^2$$

Substituting:

$$A = \sum_{i=1}^{a} n_i \left[\sigma_\tau^2 - \frac{2n_i}{N}\sigma_\tau^2 + \frac{\sum_k n_k^2}{N^2}\sigma_\tau^2\right]$$

$$= \sigma_\tau^2 \left[N - \frac{2\sum_{i=1}^a n_i^2}{N} + \frac{\sum_{i=1}^a n_i^2}{N} \right]$$

$$= \sigma_\tau^2 \left[N - \frac{\sum_{i=1}^a n_i^2}{N} \right]$$

$$= \sigma_\tau^2 \cdot \frac{N^2 - \sum_{i=1}^a n_i^2}{N}$$

Combining,

$$E(\text{SST}) = (a-1)\sigma^2 + \frac{N^2 - \sum_{i=1}^a n_i^2}{N} \sigma_\tau^2$$

Divided by $a-1$.

$$E(\text{MST}) = \frac{E(\text{SST})}{a-1} = \sigma^2 + C\sigma_\tau^2, \quad \text{where } C = \frac{N^2 - \sum_{i=1}^a n_i^2}{N(a-1)}$$

Part (b): Show that under $H_0 : \sigma_\tau^2/\sigma^2 = \gamma$, $\dfrac{\textbf{MST}}{\textbf{MSE}(C\gamma+1)} \sim F_{a-1,\,N-a}$

Since $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$, the within-group residuals $y_{ij} - \bar{y}_{i\cdot}$ depend only on the ε_{ij}'s. The standard result gives:

$$\frac{\text{SSE}}{\sigma^2} = \frac{(N-a)\,\text{MSE}}{\sigma^2} \sim \chi^2_{N-a}$$

Now, each treatment mean satisfies:

$$\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot} \sim N\left(\mu,\ \sigma_\tau^2 + \frac{\sigma^2}{n_i}\right)$$

Under $H_0 : \sigma_\tau^2/\sigma^2 = \gamma$, we have $\sigma_\tau^2 = \gamma\sigma^2$, so:

$$\text{Var}(\bar{y}_{i\cdot}) = \gamma\sigma^2 + \frac{\sigma^2}{n_i} = \sigma^2\left(\gamma + \frac{1}{n_i}\right)$$

The treatment sum of squares is a quadratic form in the $\bar{y}_{i\cdot}$'s. Using the result from Part (a), under H_0:

$$E(\text{MST}) = \sigma^2 + C\sigma_\tau^2 = \sigma^2 + C\gamma\sigma^2 = \sigma^2(1 + C\gamma)$$

Since SST is a sum of squared normal random variables with variance $\sigma^2(1+C\gamma)$ and has $a-1$ degrees of freedom:

$$\frac{\text{SST}}{\sigma^2(1+C\gamma)} \sim \chi^2_{a-1}$$

Here, MST and MSE are independent.

Taking the ratio of the two independent scaled chi-squared variables:

$$\frac{\text{SST}/\left[\sigma^2(1+C\gamma)\right]/(a-1)}{\text{SSE}/\sigma^2/(N-a)} = \frac{\text{MST}/\left[\sigma^2(1+C\gamma)\right]}{\text{MSE}/\sigma^2} = \frac{\text{MST}}{\text{MSE}(1+C\gamma)} \sim F_{a-1,\,N-a}$$

Therefore, under $H_0 : \sigma_\tau^2/\sigma^2 = \gamma$:

$$\frac{\text{MST}}{\text{MSE}(C\gamma+1)} \sim F_{a-1,\,N-a}$$